Exercise spring container.

Q1

```java
1 usage
@SpringBootApplication
public class SpringPollApplication {

    public static void main(String[] args) { SpringApplication.run(SpringPollApplication.class, args); }


    @Bean
    public String getMessage1(){
        System.out.println("hey from message1");
        return "1";
    }
}
```

Output:

hey from massage1

reason :

it's the only method there.

Q2.

```java
1 usage
@SpringBootApplication
public class SpringPollApplication {

    public static void main(String[] args) { SpringApplication.run(SpringPollApplication.class, args); }


    @Bean
    @Qualifier("1")
    public String getMessage1(){
        System.out.println("hey from message1");
        return "1";
    }
    @Bean
    public String getMessage2(@Qualifier("1") String data ){
        System.out.println("hey from message2");
        return data ;
    }
```

Output:

hey from massage1

hey from massage2

reason:

because the second method depend in the first one with the qualifier so the firs one will run first.

Q3

```java
@Bean
@Qualifier("1")
public String getMessage1(){
    System.out.println("hey from message1");
    return "1";
}

@Bean
@Qualifier("2")
public String getMessage2(@Qualifier("3") String data ){
    System.out.println("hey from message2");
    return data;
}

@Bean
@Qualifier("3")
public String getMessage3(){
    System.out.println("hey from message3");
    return "3" ;
}
```

Output 1:

hey from massage1

hey from massage3

hey from massage2

reason:

because the first and third don’t depend on any second method but the second depend on the third.

Output 2:

hey from massage3

hey from massage1

hey from massage2

reason:

as the first output so the first and third maybe one come before another but the second will wait for the third.

Output3:

hey from massage3

hey from massage2

hey from massage1

reason:

the third will come before the second but it may all execute before first.

Q4

```java
@Bean
@Qualifier("1")
public String getMessage1(){
    System.out.println("hey from message1");
    return "1";
}

@Bean
@Qualifier("2")
public String getMessage2(@Qualifier("3") String data ){
    System.out.println("hey from message2");
    return data;
}

@Bean
@Qualifier("3")
public String getMessage3(){
    System.out.println("hey from message3");
    return "3" ;
}
```

```java
@Component
public class MainController {

    1 usage
    String data;

    public MainController(@Qualifier("1") String data){
        this.data=data;
        System.out.println("hey from Main controller");
    }

}
```

Output 1:

hey from massage1

hey from massage3

hey from massage 2

hey from main controller.

reason:

the first and third don’t depend in any method and the main controller and second depend on the first and third.

Output 2:

hey from massage3

hey from massage1

hey from main controller.

hey from massage 2

reason:

the same as first but here the third and controller execute first.

Output3:

hey from massage3

hey from massage 2

hey from massage1

hey from main controller

reason:

the third execute first then the second and the first finish then the last main controller.

Output4:

hey from massage1

hey from main controller.

hey from massage3

hey from massage2

Reason:

As the third output but in reverse the first finish first and controller.

Q5

```java
    @Bean
    @Qualifier("1")
    public String getMessage1(MainController mainController){
        System.out.println("hey from message1");
        return "1";
    }

    @Bean
    @Qualifier("2")
    public String getMessage2(@Qualifier("3") String data ){
        System.out.println("hey from message2");
        return data;
    }

    @Bean
    @Qualifier("3")
    public String getMessage3(){
        System.out.println("hey from message3");
        return "3" ;
    }
```

```java
import org.springframework.beans.factory.annotation.Qualifier;
import org.springframework.stereotype.Component;

1 usage
@Component
public class MainController {

    1 usage
    String data;

    public MainController(@Qualifier("2") String data){
        this.data=data;
        System.out.println("hey from Main controller");
    }

}
```

Output:

hey from massage 3

hey from massage 2

hey from main controller

hey from massage 1

reason:

because the third don’t depend on any method it will execute first then the second because it depend on the third the the main controller then the last the first .